

Connecting Mobile Money: Understanding the Adoption and Usage of Mobile Financial Services (MFS) by Low-Income Internal Migrant Workers in Dhaka

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Abstract

With the emergence of prominent mobile financial service providers in developing countries, it is essential to have a comprehensive understanding of how mobile money flows and trickles down, exerting a profound impact on the local economy. There is a need to understand better and identify the MFS usage pattern of low-income migrant workers and how their transactions permeate the family economy. This aspect remains largely unexplored in the existing HCI literature. To address this gap, we present the findings from our qualitative study, which analyzes the responses of $n=40$ (Male 32 and Female 8) low-income internal migrant workers in Bangladesh. Semi-structured interviews were conducted on-site at six different locations across Dhaka¹. We observe a significant correlation between users' education level, profession, and age demographics with their utilization of MFS. Furthermore, our study explores the impact of mobile money and its adoption as a mass-market product in an emerging digital economy.

CCS Concepts

• **Human-centered computing** → **Human-Computer Interaction (HCI); User Studies.**

Keywords

Bangladesh, Low-Income Internal Migrant, Mobile Financial Services, Economic Growth

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1 Introduction

Dhaka city witnesses a daily surge in internal migration from various regions across Bangladesh. Consequently, a substantial segment of the city's populace comprises low-income migrant workers, predominantly engaged in occupations such as laborers, drivers, domestic aides, building caretakers, etc. As the capital and financial hub of the nation, Dhaka's economy thrives on the hard work of its residents. Despite their significant contributions, many migrant workers struggle to support the families they left behind, often hindered by limited access to traditional financial services and technological advancements. However, with the advent of the digital economy and the widespread adoption of mobile technology, there has been a remarkable surge in mobile phone usage nationwide. This trend has created a significant opportunity for the emergence of local Mobile Financial Services (MFS) providers in the consumer space. Survey data shows that MFS users increased from 21% in 2017 to 29% in 2021 [11].

Mobile banking has started to become the norm. We observe that services traditionally offered by banks and financial institutions, once perceived as inaccessible due to various entry barriers and logistical constraints, are now readily available to the general population [2]. People who previously thought "Banks are not for me" now utilize the benefits of digital transactions [13]. With the country's emergence and the rapid rise of local MFS companies (bKash, Nagad, Upaybd, Rocket), mobile money is now a booming business [9]. The advancements in digital technologies have opened up unprecedented opportunities to uplift individuals in the lower-income demographic, offering them access to resources and services previously beyond their reach. As cashless transactions gain prominence, accountability increases, and financial transactions become more transparent. Better understanding the unique challenges faced by individuals managing a low income creates opportunities for innovative digital services that extend sophisticated financial practices to this population demographic.

Understanding the impact of vulnerable and underserved populations is key to recognizing their role in the national GDP. Research is needed to examine the adoption of Mobile Financial Services (MFS)

in daily life, analyzing usage patterns and the challenges faced by our target group. Our study focuses on MFS use among migrant workers in Dhaka. Since MFS requires basic technical and mobile literacy, it's crucial to identify pain points to meet their needs and improve their quality of life. The study aims to align MFS with the needs of internal migrant workers, focusing on financial inclusion, usability, and infrastructure to highlight the benefits of a digital economy. It specifically targets low-income internal migrant workers in MFS adoption. Previous findings show that lower-income workers face discrimination in traditional banking systems and establishments [13]. Marginalized individuals historically faced barriers in accessing formal financial institutions, including challenges in managing savings, obtaining loans, and reliance on cash-based transactions. Transition into Mobile Financial Services (MFS) has enabled them to overcome many of these hurdles [17].

Nevertheless, there remains a scarcity of impactful research focusing on these marginalized groups despite their significant contribution to the urban workforce. From previous studies, we get an idea of the impact of mobile money on household income. However, very few studies analyzed the large-scale cashless cash flow of mobile money through these digital transactions [38] and aimed to improve the cashless systems integration into the existing societal framework of the global south [34]. Exploring these aspects, among other strategies, can provide valuable insights into users' experience and utilization of MFS. Contributing to improving services that specifically ensure the needs of these low-income and low-literate migrant workers, who represent a substantial portion of the country's working population.

Our research also seeks to understand the factors influencing the adoption of MFS services among the migrant workforce and its usability as a mass-market product. In particular, we look for the answer to the following research questions:

RQ-1 : *What are the perceived benefits for migrant workers that influence the general adoption of mobile financial services over already existing conventional financial infrastructure?*

RQ-2 : *How does the use of MFS vary among different demographic groups within internal migrant workers? More specifically, does it reflect and/or reinforce existing socioeconomic inequalities?*

To answer these questions, we conducted qualitative interviews with 40 participants. **RQ1** delves into the perspective of migrant workers and their acquaintance with mobile financial services to elucidate MFS services' impact on their daily lives and the perceived problems they face in utilizing them. Besides, it addresses the broader landscape by identifying factors influencing the adoption of MFS over traditional financial systems. Contributing to a better understanding of the evolving financial preferences of migrant workers. **RQ2** aims to uncover potential disparities in MFS usage among different demographic groups of internal migrant workers, providing valuable insights into societal inequalities. It explores how social inequalities are reflected in MFS platforms and how these services may reinforce existing social norms. Investigating these research questions is essential for gaining a comprehensive understanding of MFS usage dynamics among internal migrant workers.

In addressing the research questions given above, we make two contributions to the HCI4D/ICT4D communities as follows:

- Our analysis of MFS adoption by low-income internal migrant workers in Dhaka provides insight into how mobile money usage shifts based on user demographics. By reviewing interview discussions, we identified patterns explaining the challenges faced by this population in traditional financial systems.
- Using thematic analysis, we examined key factors affecting migrant workers' experiences. Our qualitative research evaluates the role of digital financial services in developing economies and the impact of MFS technologies. We found that socioeconomic factors, such as gender inequality and job diversity, shape users' daily behaviors.

2 Related Work

A significant portion of prior research conducted in the domain of mobile financial services has focused on how to successfully implement and deploy mobile money services and how the acceptance of these services into the general population significantly impacts the digitization of the economy of developing nations. However, there has been a lack of qualitative studies that have identified the key factors influencing the adoption of MFS and the circulation of mobile money in daily transactions. Given the vast scope of research on the financial demographics of low-income workers, our literature review concentrated on investigating the correlation between Mobile Financial Services (MFS) and the lifestyle of migrant workers.

2.1 Digitization and Mobile-money

Previous empirical studies examining the correlation between the digitization of an economy and the widespread adoption of digital financial services have predominantly concentrated on financial inclusion [4, 6, 14, 18], mobile banking [8, 14, 17, 32], product usability [12, 31, 33] and service infrastructure [15, 29, 31]. Existing knowledge has highlighted that increased affordability and penetration of internet-enabled smartphones among low-income groups directly correlates with the digital revolution of an emergent economy [12]. With an uptick in mobile phone usage, 79% of adults now own mobile phones [11] with almost 190 million telecom subscribers in Bangladesh [22]. MFS provides a channel to interact with various financial institutions, encompassing mobile payment, mobile money transfer, and mobile remittance services via a mobile device [18]. Runaway success of M-Pesa in Africa has enabled the emulation of similar money-mobile models all over the world [19]. Numerous studies also analyze the factors behind the successful implementation of mobile money [8]. The advantages of mobile money include "mobility" and "flexibility" by facilitating a low barrier to entry [24]. As reflected in GFD reports, 45% of adults in Bangladesh have made/received digital payments at least once in their lifetime [11]. This transition to a digital economy was easier for those already comfortable with non-cash transactions [31].

2.2 Challenges of Widespread MFS Adoption

Studies [13] describe that individuals with lower socioeconomic status in developing nations feel that formal banks do not offer services tailored to their needs. The socioeconomic stigma of not

being wealthy enough to engage with financial services significantly impedes the adoption of better alternatives. Author Simone [8] describes the widespread adoption of mobile phones as enabling the development of mobile money systems, which deliver financial services through the mobile phone network. While minimizing the risks [16], mobile money services have also become a viable means of access to financial services by people who initially thought “Banks are not for me” [13].

Uneducated, poverty-stricken, marginalized communities suffer the most when adapting to digital transitions [12]. Nevertheless, the association between poverty and limited formal education raises concerns about whether providing digital access to intricate financial services alone can narrow the existing gap of adoption [24, 25]. Recent studies [18] have noted that providing financial services at affordable costs to the disadvantaged and low-income segments of societies is considered to be one of the major enablers of economic development. With 60 million unbanked adults in Bangladesh and 1.4 billion unbanked adults worldwide [11], individuals lacking any formal bank accounts predominantly engage in cash-based economies, constraining their ability to secure loans, manage savings, or avail themselves of any financial services [17]. In contrast, only a 1% increase in the population using mobile devices increases general household income by roughly 0.3% points [10]. However, to benefit the majority, mobile money solutions must scale up and strive to serve a critical mass of users. [7].

2.3 User Inclusivity

Studies done in India show that through MFS, a large percentage of Indian workers can be made financially inclusive in the national economy, which has been an elusive target for decades [19]. In this new age, the consequence of uneven adoption of new technologies may reinforce old inequalities [31]. Although a small percentage of Indians living in cities used non-cash transactions, most of them lacked working bank accounts [12, 19, 31]. The factors influencing the diverse patterns of adoption are multifaceted [20, 28, 31]. In a comprehensive study of five mobile banking services, Medhi et al. [26] discovered that the adoption rates among individuals with low literacy and income levels depend on several critical factors, including the type of household, specific mobile banking services adopted, the adoption rate, usage frequency, and perceived ease of use.

2.4 Migrant Workers and HCI

In the particular case of researching migrant workers’ transactions of domestic and international remittance [27], it has been identified that mobile money has enormous potential from both client and revenue standpoints [19]. Migration typically arises from the pursuit of improved quality of life, as individuals seek opportunities for enhanced living standards and overall well-being [35–37]. Alam et al. [5] describe the disproportionate differences in opportunities between the metropolitan and other cities as the key reason behind internal migration within a country. Research highlighting Bangladesh finds that 50% of the migrating demographic were in the age group of 20–30 years, and the highest proportion were migrating to Dhaka city [3]. Similar notations also demonstrate that the migration of the working-age population has mainly had

positive effects on the economy [1, 5]. Internal migration has often been treated as both hero and villain of national development [5]. Based on cost-benefit analysis, young individuals still journey into cities to minimize family risk of poverty and hope for a better future [5, 35]. Everyday spending behaviors and experiences usually depend on the target population. Most migrating workers send a portion of their income back home to their family, investing in household items and livestock [3, 36].

In recent years, advancements in digital technology have remained synonymous with the overall development goals of a country [14, 28, 31]. Entry into the cashless economy means money can now travel with you anywhere, whenever needed [8, 18]. MFS customers deposit and withdraw cash from their accounts through a network of retailers or registered mobile money agents. Various mobile platforms, including USSD², SMS, and smartphone applications, serve as accessible channels for individuals to manage these transactions on their mobile devices [8]. Now, migrant workers must confront the challenges of understanding smartphones and utilizing mobile financial services themselves [12, 13, 29]. Empowering them to pursue their priorities is crucial when crafting designs for marginalized communities. The HCI community must play an active role in understanding the motivations behind mobile money transactions, particularly in the form of domestic remittances and associated micro-services [29] aligning with the needs of migrant workers.

3 Methodology

Access to finance is essential for economic growth. Mobile financial services refers to users conducting financial transactions with their own devices through digital mediums. Our study is structured towards an understanding of these transactions and the flow of capital among users. To gain knowledge about how its usage varies from person to person. As we primarily focus on internal migrant workers who came to Dhaka seeking better financial opportunities and are separated from their families back home. To understand MFS adoption of these users, we needed to explore the dynamics of prominent providers like bKash, Nagad, Rocket, Upay, etc in the local market. Figure 1 shows us the current market share of these providers. Where bKash and Nagad dominate the sector, holding over 85% in terms of transactions [23].

We primarily employed qualitative research methods with the inclusion of some quantitative data collected while conducting in-person semi-structured interviews with participants in multiple locations across Dhaka city. Pre-constrained assumptions were made regarding participant sampling with a viewpoint to explore in-context responses from our target population and the potential impact of our collected interview data following our research goal of understanding the utilization of MFS by individuals of different ages and professions. These interviews also helped us understand the potential benefits and challenges faced by the low-income, low-educated population.

3.1 Sample Design and Data collection

A stratified sampling method was used to select the study population. With the consent of all participants, individual interviews

²Unstructured Supplementary Service Data

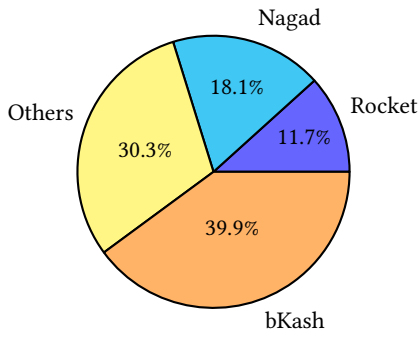


Figure 1: Marketshare of local Mobile Financial Services in Bangladesh

ranging from 20 minutes to 25 minutes each were conducted using a researcher-made questionnaire set consisting of 25 interview questions, which include multiple-choice questions, open/closed-ended questions to document qualitative data. Selected multiple-choice questions and Likert scales were also used to document quantitative data regarding approximate transaction amount and user satisfaction score. All the interviews were primarily conducted in the Bangla language to understand the interviewees better. All interviews were then translated and transcribed into English. No monetary compensation was provided for the participants.

As shown in table 1, the interview questions are divided into 9 sections, each exploring the user's understanding of different data elements. The interviews begin by collecting basic background information like name, age, and occupation, helping to group participants by demographic categories. Since our focus is on migrant workers in Dhaka, it's important to gather data on their motivation and time spent in the city. Key questions about their MFS provider and daily activities help gather critical insights into their overall MFS experience. Open-ended questions were used to encourage in-depth discussions, and the questionnaire was validated for reliability before the interviews.

3.1.1 Data Analysis Methods. The purpose of these questions is to bring forth user data and create engagement queries from the participants. Focusing on who uses these services, why they are used, and how they use them. The questionnaire set was carefully constructed to ensure that characteristics appropriate to our research questions were represented in the sample. Observations of these interviews and user habits of mobile financial services in the study population highlight the impact they have in their lives.

We analyzed qualitative data using structured documentation, coding, and thematic analysis. Preliminary codes were assigned to capture the meaning of participants' responses. Our research focuses on understanding specific features and usage patterns of MFS technology among a particular user base. Open-ended questions were crucial for exploring differences in user behavior. The authors coded the data to identify patterns and themes, grouping them into broader categories across interviews. Each coder created a codebook and developed key themes. Merging different perspectives, we introduced major nested themes that shaped our findings. We then reviewed the relevance of these themes to our data and research

goals. Table 2 shows the visual representation of the identified parent themes and their corresponding nested themes. Summary statistics for quantitative data are also included and discussed in the findings.

3.2 Ethical Considerations

In conducting this research, utmost care has been taken in strict adherence to ethical principles. Participants were treated with respect, and their anonymity was strictly maintained. Informed consent was obtained from all participants. We prioritized fairness, cultural sensitivity, and inclusivity throughout the research process. Moreover, we maintained ethical research standards by securing approval from the Institutional Review Board (IRB). This study is committed to ensuring the well-being and privacy of all involved.

4 Findings

We used thematic analysis to examine the qualitative data from our interviews. Preliminary codes were applied to capture the significance of participants' responses. We followed a deductive coding approach, as our research aims to understand specific features and patterns of MFS usage by a particular user group. Open-ended questions were used to gather in-depth insights through meaningful discussions. We searched for patterns in the codes across interviews and grouped them into broader categories. Each coder created a codebook and identified key themes. Combining perspectives, we identified major nested themes that informed our findings. We then reviewed the themes for relevance to our data and research goals. Table 2 provides a visual representation of the parent and nested themes.

In this section, we describe the results of our thematic analysis and interview data and answer the aforementioned research questions based on these results.

4.1 RQ1: General Adoption and Benefits of Using Mobile Financial Services

Our interview data shows that low-income migrant workers move to cities like Dhaka mainly for better financial opportunities. However, due to limited knowledge of traditional financial institutions, they often rely on third-party cash transactions to send money home, sometimes involving someone physically delivering the cash and taking a fee. These complex methods increase the risk of theft, fraud, or loss during transfers. Our findings highlight the impact of MFS adoption, with 37 out of 40 participants having accounts with at least one local provider. Users switching to MFS associate mobile money with quick access to their funds. However, marginalized groups may struggle to fully understand the benefits of mobile money services and often continue relying on familiar methods or depend on others to manage their accounts.

4.1.1 MFS Adoption Mediums. The usefulness of MFS depends on factors such as the service's availability, reliability, and security, as well as the range and quality of the features and functions offered. We found that 26 out of 37 participants who use MFS services use their smartphones to access the services. Our collected data also reflects the growing adoption of smartphones in the lower-class worker demographic. Figure 2 shows that almost 70% of our

Table 1: Summary of interview questionnaire

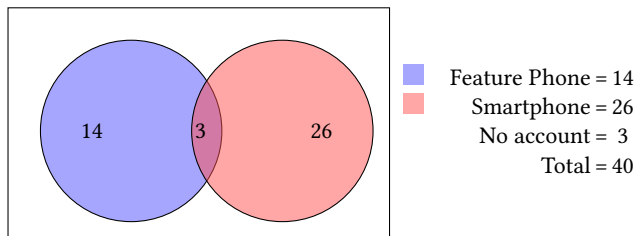
Section no.	Section	Question no.	Response format
Section 01	Personal Information & Demographics	1-8	Simple text and multiple-choice
Section 02	Migration to Dhaka City	9-10	Simple text and multiple-choice
Section 03	Information about mobile device usage	11-12	Multiple-choice
Section 04	Mobile Banking services	13-15	Binary-choice and multiple-choice
Section 05	External mediums of assistance	16-17	Multiple-choice
Section 06	Type and amount of MFS transactions	18-20	Multiple-choice
Section 07	MFS in comparison with traditional banking services	21	Simple text
Section 08	Challenges of sending money back home	22	Simple text
Section 09	User feedback & benefits of MFS services	23-25	Multiple-choice

Table 2: Thematic analysis of interview data

Parent Theme	Nested Themes
MFS Adoption in stages	User hesitancy, easier transaction, external dependency
Use of services and benefits	Transaction amount, income, user convenience
User pattern and cash flow demographic	Education, age demographic, profession, living standard
Obstacles in utilizing MFS services	Insecurity, mass market adoption, user accessibility, negative experiences
Observing inequalities	Social norms observed, work recognition, gender inequality

Table 3: Participant Occupation and Gender Index

Participant Occupation	Male	Female	Count
Professional/Freelance Driver	7	0	7
Day labourer	4	1	6
Rickshaw puller	4	0	4
Domestic help	1	6	7
Security Guard	8	0	8
Hostel attendant	0	1	1
Building Caretaker	6	0	6
Barber	2	0	1
Total Count	32	8	40

**Figure 2: Distribution of Smartphone and Feature Phone (Non-Smartphone) Users in the Study Population.**

study population use smartphones as their primary mobile devices. However, feature phone or non-smartphone use also tends to be higher in older age groups. Feature phone users are only subjected to the use of USSD for their daily mobile money transactions. Some participants stated that using smartphones at this age is too tricky,

and they are content using their feature phones. Consequently, they are only introduced to key MFS services like sending money, mobile balance top-up, and payment.

However, advanced services like savings, loans, insurance, and other payment shopping benefits can not be easily accessible through USSD requests. The development of these services prioritizes digital adoption through a robust mobile app. In such cases, migrant workers may depend on others, such as intermediaries, agents, or peers, to help them use their accounts. This reliance is subject to factors such as lack of access to mobile devices, network coverage, electricity, or lack of confidence to operate the technology. This affects the service cost, convenience, and security, as well as the privacy and autonomy of the users. While analyzing the answers of the participants who shared their stories about adopting mobile money services, our research illuminates the theory that MFS adoption happens in stages and that various external factors directly or indirectly affect the general adoption of users.

4.1.2 Challenges of MFS Adoption. Our analysis highlighted the importance of service reliability, especially for institutions and banks associated with MFS providers, in building trust. Users familiar with traditional financial services and those who only dealt with hard cash had varied experiences. Migrant workers, seeking better opportunities, face difficulties managing their income. They want a secure way to send money to their families, ensuring their efforts are recognized and their loved ones cared for. A fast, cost-effective system with minimal risks of loss or delay is crucial for their peace of mind.

With the rise of digital services, financial inclusivity has become vital, especially for migrant workers. Despite the advantages of MFS, like lower costs and faster transactions, adoption among these workers remains a challenge. Many hesitate due to trust issues, lack of awareness, or unfamiliarity with the technology. Barriers such

as language, literacy, and regulatory procedures further complicate adoption. We interviewed one participant who firmly believed in cash money and did not want any more chances with mobile money with his hard-earned cash as he had already faced difficulties in recovering his bKash account after misplacing his mobile phone.

4.1.3 Services and Benefits. In our interviews, 33 respondents reported using bKash, while 3 exclusively used Nagad. Additionally, 17 participants used both services. This can be compared with the overall market share of these local MFS providers shown in figure 1. These findings highlight the strong market presence of these two companies, reinforcing their dominance in Bangladesh's growing Mobile Financial Service sector. The benefits of mobile money services go beyond convenience, offering greater financial access, security, and faster transactions. This has now become standard practice, introducing users to safer money transfer methods and easier access to their funds. Service providers have opened up new possibilities for users to manage their money instantly.

One female participant shared how MFS significantly transformed her life for the better. A widowed mother with two daughters who live with their grandmother in the countryside. She has been working as a domestic helper in Dhaka for 12 years. She explained how her illiterate mother's lack of knowledge led their family to fall victim to financial scams in the past. However, when she was introduced to MFS through a local agent, it gave her relief and security, ensuring her money safely reached her family. Now, with her daughters learning to handle basic MFS functions, they can get hold of her money when needed, without involving help from anyone else. In the past, traditional banking processes required basic literacy, which she and her mother lacked, forcing them to rely on brokers. With MFS, she feels more in control of her family's expenses, giving her a sense of independence and confidence.

Additionally, during our conversation with one male participant, who moved to the city at 16 years old to start working as a cook. He expressed how mobile money services have seamlessly transformed his life, subduing the stress associated with traditional banking methods and unauthorized transactions. Being reasonably educated, he found using MFS services with his smartphone quite convenient compared to the weekly routine of going to the bank and waiting in line. He also described the relief he felt when the service agent only needed his national identity (NID) card and mobile number for the initial registration process.

Benefits such as a lower barrier of entry, a lower learning curve for the low-literate marginalized population, and the ease of access to basic financial services through mobile devices are not just conveniences. They are tools of empowerment, giving these individuals more control over their money and how it is used.

4.2 RQ2: Observing Factors of Influence Across Different Age and Profession Demographic, Presence of Socioeconomic Inequalities

Our study findings reveal that our participants face various technological and socioeconomic challenges. However, we also identified key factors driving the growing use of Mobile Financial Services (MFS) among the study population. Demographic features such as

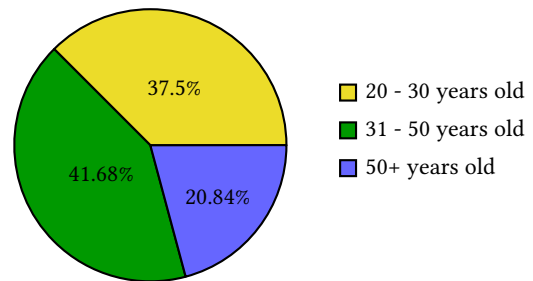


Figure 3: Age distribution of the study population. The pie chart illustrates the percentage distribution across different age groups: 20-30 years old, 30-50 years old, and 50+ years old.

education, age, profession, and current living standards significantly influence the adoption and use of these technologies.

During the interviews, we observed that individuals over 40 often avoid using smartphones in their daily lives. Many feel they lack the capability to manage their own MFS accounts. Figure 3 shows the age distribution of our study, with participants from various age groups. Interestingly, some respondents mentioned they had smartphones but relied on more tech-savvy family members for MFS tasks. Some had negative past experiences with MFS, finding it too complicated. This suggests that the fear of financial mistakes outweighs the potential convenience. While smartphone usage rates were similar across age groups, older individuals are more protective of their money and less likely to use MFS for personal finances.

Our open-ended interview questions revealed that education plays a crucial role in adapting to new technologies. Those with higher education levels are more likely to adopt technology. All participants with at least a high school education used smartphones, all of whom used MFS services. Their education and ability to understand instruction languages gave them the confidence to access these financial services. In contrast, the uneducated working class showed little interest in exploring MFS options, struggling to engage with either traditional or modern financial systems.

People's professions and living situations also have an impact on how people face obstacles in utilizing services. Table 3 shows us the participant's occupation and gender index encompassing 6 different professions. People who are dependent on MFS transactions for their daily earnings, like freelance drivers, have a lot more time to interact and slowly get used to the features that mobile money services provide to their full potential. Meanwhile, domestic helpers may only stick to the basic functionalities they learn. Most people who work on-site or in-house jobs, such as domestic help, building caretakers, and attendants, only regard MFS services as a medium for sending and receiving money with their relatives. They are concerned about the convenience of the process but lack the encouragement to dive a little deeper into what other services are available.

Additionally, the participants' background data revealed that users' family dynamics also play a pivotal role in MFS utilization

in daily life. In several instances, direct kinship relationships exerted a significant influence on the decision-making process about technology adoption. Notably, older participants frequently engaged in discussions regarding their financial transactions, seeking validation and confirmation from their inner circles, which were comprised of close friends and family members. This tendency underscores the social dimensions that intertwine with technological assimilation among the older demographic.

4.2.1 User Pattern and Cash Flow Demographic. The amount of financial transactions also emerges as a important determinant in the stratification of specific demographic cohorts, such as younger working adults (ages 18 -30) who tend to send portions of their monthly income back through mobile money services once or twice a month. Most of these individuals have begun working in Dhaka within the last decade.

Two participants, who were security guards by profession, proclaimed that they came to Dhaka to ease their families' burdens. One's father fell ill, and another was burdened by heavy agricultural loans. Losses in multiple seasonal harvests forced his father to send him to Dhaka in search of work. The majority of the middle-aged working demographic in our study population maintain their own families back home. They are the family's breadwinners. Hence, they sometimes need to use the MFS send money function multiple times weekly.

Our interviewees stated that digital money services especially come in handy with utility bill payments. Even rural utility service providers have now embraced the usage of mobile money transactions. Some MFS providers have made it easier by directly partnering with these services to attract more users to their platforms. These changes also influence long-standing social norms.

4.2.2 18-30 Years Old. Our study focused on young adults aged 18 to 30, who moved from rural areas to Dhaka in the last 5 to 10 years to improve their families' living standards. This group is technologically adept and relies heavily on smartphones to access MFS services, using apps to track transactions. Some participants reported using MFS more than 20 times a month, not only for remittances but also for other functionalities.

We interviewed 9 participants, mostly male freelance motorcycle drivers, who earned through digital payments. Many had no ties to formal banking, with their wages accumulating only in MFS accounts, underscoring the service's reliability. Interestingly, their monthly transaction amounts did not differ much from older workers, but the usage patterns, cash flow, and goals for the funds varied.

4.2.3 31-50 Years Old. This group is more established in Dhaka, having worked there longer and gained better control over their income. Many said they support their own families and often think about how to use their earnings from Dhaka more efficiently. Most can operate MFS on their own, though some still need help from peers or telecom shops. Many use both feature phones and smartphones for daily tasks.

Although the total transaction amounts don't vary much across jobs, changes in cash flow patterns are noticeable. For example, one male barber makes small transactions several times a day. Whereas one male security guard sends larger sums less frequently. Both

send less than 15,000 taka home each month. They also use MFS for utility payments to meet family needs. Our interviews show that these workers focus on maintaining stability for their families, using the money for things like education, food, and farming. This highlights their influence over household decisions, with family members aware that their loved ones are working hard in the city to improve their living standards.

4.2.4 50+ Years Old: People in this category mostly moved to Dhaka city over 20 years ago. According to our observations, this subset of people is more likely to avoid using the complexity of smartphones. While a portion of them may use smartphones, most of the study population prefers to use non-smartphones. Most of them have been the only breadwinner for their family for many decades. Most of their kin have moved out of the family homes. Some now live in the city with their children. However, it is predominantly clear that this working-class demographic is the most vulnerable to financial scams. Due to their lower technical literacy, most participants fully depended on external assistance to manage their MFS needs. The local telecom shops have an important role in assisting this older group of people. The cash flow from the income generated is now used for development needs. The participants we observed conveyed their wish to build or repair their house back home, buy cattle, buy medication, and invest in agricultural production.

4.2.5 Male Dominance and Urban Migration. While Dhaka city is characterized by the significant presence of male migrant workers, it is essential to acknowledge the diverse demographics within the urban landscape, including women, who contribute to the city's workforce and societal fabric. Traditionally, female migrant workers are thought to be seen only in the sphere of the domestic help workforce. Table 3 also exhibits that out of the 7 domestic help workers we interviewed, 6 are female. The male participant reportedly earns a higher salary than others, reinforcing gender stereotypes.

Differences in income in different fields of work create potential variations in mobile financial services (MFS) usage between male and female migrant workers, revealing the complex changes in the growing world of labor migration. Female migrant workers, often employed in domestic roles, may use MFS for remittances and managing their finances. This group has specific financial needs that MFS can address. Where ease of usability is identified as their primary concern while engaging with financial services in any form, understanding these gender-specific patterns is crucial for tailoring financial services to address the diverse needs of both male and female migrants, enabling coherence in mobile money services. MFS can be a crucial tool for financial inclusion, providing services to individuals who might not have had access to formal banking in their rural hometowns.

4.2.6 Observed Inequalities and Challenges. The study delves into the socioeconomic implications of MFS adoption, emphasizing the impact on family dynamics, social norms, and financial independence among migrant workers. It recognizes the role of family networks in influencing technology adoption decisions, particularly among older demographics. The age-specific analysis provides valuable insights into the varying needs and preferences of different

age groups. In our small-scale study, 20% of our interviews were with women. 75% of these female participants are in the sphere of domestic help workforce.

While digital payments can contribute to a country's development, the experiences of individuals vary widely. People from lower economic backgrounds often face more challenges. One participant shared an experience of not being able to cash out a small amount due to an uncooperative MFS agent, which may stem from agents prioritizing profit. Women, especially working-class women, face additional challenges, like needing to visit agent shops multiple times a month, often in difficult social settings.

5 DISCUSSION

Our qualitative study examines MFS utilization of low-income internal migrant workers, its prospects of mass-scale adoption, and identifying obstructions the study population faces through its use case. Subsequently, trying to make sense of the impact of this massive flow of digital capital in a growing economy. Our analysis dives into the intricate dynamics of mobile financial services (MFS) adoption among migrant workers in Dhaka, offering insight into exploring their challenges, preferences, and usage patterns. The study identifies key factors influencing the adoption of MFS over conventional financial infrastructures and investigates potential disparities in usage across demographic groups. The study highlights the impact of MFS on the daily lives of migrant workers, emphasizing the quest for a reliable, quick, and cost-effective medium for managing and sending their hard-earned money back home. Additionally, the research sheds light on the hurdles users face, such as trust issues, awareness, and technological familiarity, while illustrating the challenges individuals face in transitioning from traditional banking to mobile money services. Three major trends emerged from our interviews.

5.1 Role as a Mass-market Product

Being a profit-driven mass-market product, MFS providers seemingly focus better on capitalizing on the demographic that earns them greater revenue. However, to reach a greater audience, it has to take strong initiative and increase accessibility to low-literate people. Speculative market research should be conducted on specific categories of people to understand the overlaying dynamics behind their usage patterns. Low-income migrant workers can depend on these services' security infrastructure to ensure their hard-earned money is being received by their loved ones effortlessly. This transition to mobile money will take time, especially with a marginalized and ill-catered demographic like low-income migrant workers, who rely on daily income for their daily expenses. For those who are hesitant to explore and engage with digital money transactions, a strong need for use cases has to be presented. It must also be emphasized that mobile payments will co-exist with (not replace) cash [10]. Some users may feel overwhelmed by the recent inclusion into the mainstream finance sectors, where the biggest vendors now have over 70 million users [11, 21], and the total market generates over 22 Billion TK in transactions [4, 5].

5.2 Perceived Advantages and the Challenges Associated

As a study narrows down to a very specific demographic, polarities are hard to distinguish between the same category of participants. Common traits such as accessibility of services to low-literate people and the challenges associated with adopting a new technology after being neglected by existing traditional banking systems over the years are observable in this study. The study also highlights people's satisfaction and relief over current MFS services that are suited to their use case. People using MFS services now have more freedom and transparency over their own money and how it proportionately benefits their loved ones. Common benefits encompass the significance of financial inclusivity, particularly in the context of migrant workers who seek seamless, secure, and accessible financial solutions. The study reveals a considerable adoption of MFS among the interviewed participants, with dominant players like bKash, Nagad, Rocket, and other small players playing a pivotal role in shaping the mobile financial landscape. Notably, the research highlights the prevalence of smartphones among younger migrant workers, indicating a potential shift towards digital payment methods and the need for tailored solutions for different age groups.

5.3 User Satisfaction of Target Population

The study delves into the socioeconomic implications of MFS adoption, emphasizing the impact on family dynamics, social norms, and financial independence among migrant workers. It recognizes the role of family networks in influencing technology adoption decisions, particularly among older demographics. The age-specific analysis provides valuable insights into the varying needs and preferences of different age groups, emphasizing the need for targeted strategies to address the diverse requirements of migrant workers. The study also identifies the increasing trend of MFS usage for utility bill payments, showcasing the expanding role of these services beyond remittances. One notable aspect is the dependence observed among certain demographics on intermediaries and local telecom shops, emphasizing the need for user-friendly interfaces and support for those who are less technologically literate. A more forgiving system for transactions and an explicit automated voiced-based transaction confirmation system can be implemented [30] to cater to their ease of use. This study also highlights the importance of building confidence between service providers and users, especially among older individuals who may resist technological change.

5.4 Limitations and Future work

Considering the breadth of our analysis, we recognize certain limitations that pave the way for future exploration. Our research primarily concentrates on low-income migrant workers presently employed in Dhaka. This deliberate focus has led to a small-scale data collection process with a constrained number of participants, thereby excluding individuals from diverse professions. Due to resource constraints, conducting a large-scale study with a broader sample size was unfeasible.

Moving forward, there exists an opportunity to delve deeper into understanding the usability constraints of our target population.

Future research endeavors could pivot towards the application of inclusive design principles, offering tailored solutions to address the insights gained from our study. This approach not only enhances product adoption but also ensures greater inclusivity and usability across diverse user groups.

6 CONCLUSION

In conclusion, our examination of mobile financial services (MFS) adoption, particularly among low-income internal migrant workers, offers valuable insights into the dynamics of mobile money usage and its responsiveness to user demographics. In the era of digitization, there is a growing emphasis on expanding access to MFS in emerging economies. It is imperative to prioritize the inclusion of low-literate, lower-income workers, who form a vital part of Dhaka city's economic fabric. By recognizing their needs and preferences, MFS products and services can be tailored to better serve this demographic.

Over recent years, mobile financial services have emerged as a cornerstone of both rural and urban economies, facilitating significant cash flows. Understanding the deep-rooted connection between MFS and users enables us to address the limitations and barriers identified in this qualitative study. Moving forward, integrating these insights into the planning and design of MFS offerings will contribute to greater accessibility and utility for all users, ensuring the distribution of the benefits of digitization throughout society.

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